

Arrays and Structures

Pattern Matching

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Outline

- 1 The String ADT
- 2 The String Matching Problem
- 3 The Naïve Algorithm
- 4 Knuth-Morris-Pratt (KMP) Algorithm



String ADT

- A string is:
 - objects: a finite sequence of zero or more characters.
- functions: for all strings s, t , and $i, j, m \in \mathbb{Z}^+$:
 - String $\text{Null}(m) ::=$ return a string whose maximum length is m characters, but is initially set to NULL. We write NULL as "".
 - Integer $\text{Compare}(s, t) ::=$ if s equals t return 0 else if s precedes t return -1 else return $+1$
 - Boolean $\text{IsNull}(s) ::=$ if $(\text{Compare}(s, \text{NULL}) \neq 0)$ returns FALSE else return TRUE (equivalent to `s.empty()` in C++).
 - Integer $\text{Length}(s) ::=$ return the number of characters in s (equivalent to `s.size()` or `s.length()` in C++).



String ADT (functions contd.)

- $\text{String Concat}(s, t) ::=$ return a string whose elements are those of s followed by those of t else return s .
- $\text{String Substr}(s, i, j) ::=$ if $((j > 0) \ \&\& \ (i + j - 1) < \text{Length}(s))$ return the string containing the characters of s at positions $i, i + 1, \dots, i + j - 1$. else return NULL.
- $\text{void StrInsert}(s, t, i) ::=$ insert string t at position i of s .



String Insertion (code)

```
void strInsert(char *s, char *t, int i) {  
    /* insert string t into string s at position i */  
    char string[MAX_SIZE], *temp = string;  
  
    if (i < 0 && i > strlen(s)) {  
        fprintf(stderr, "Position is out of bounds \n");  
        exit(1);  
    }  
    if (!strlen(s))  
        strcpy(s, t);  
    else if (strlen(t)) {  
        strncpy(temp, s, i); // Copy the first i characters of s to temp  
        temp[i] = '\0'; // complete a string with '\0'  
        strcat(temp, t);  
        strcat(temp, (s + i)); // concatenate the rest characters of s to temp  
        strcpy(s, temp);  
    }  
}
```



String Matching

String Matching

- **Input:** Two strings s and p of sizes n and m respectively.
- **Output:** Determine if p is a substring of s . Return the starting position of s if it is and -1 otherwise.



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- **Substring:** a contiguous sequence of characters within a string.
- For example,
 cgta is a substring of **acgtacct**.
 acgg is NOT a substring of **acgtacct**.



Outline

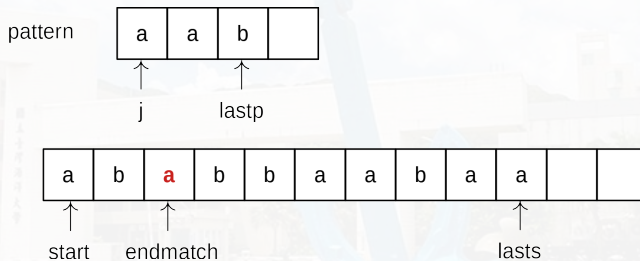
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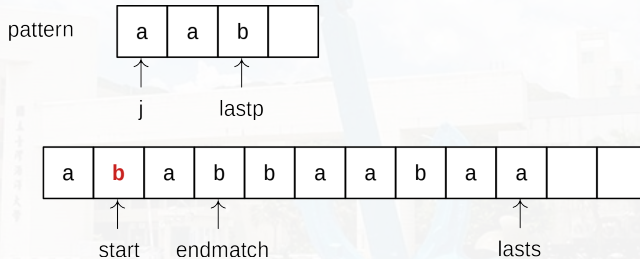


- Why do we match the last character of the pattern first?

```
int naiveFind(char *string, char *pat) {  
    /* match the last character of pattern first,  
       and then match from the beginning */  
    int i, j, start = 0;  
    int lasts  = strlen(string) - 1;  
    int lastp  = strlen(pat)    - 1;  
    int endmatch = lastp;  
  
    for (i = 0; endmatch <= lasts; endmatch++, start++) {  
        if (string[endmatch] == pat[lastp])  
            for (j = 0, i = start; j < lastp && string[i] == pat[j]; i++, j++)  
                ; // empty statement: advance while matching  
        if (j == lastp)  
            return start; /* successful */  
    }  
    return -1;  
}
```



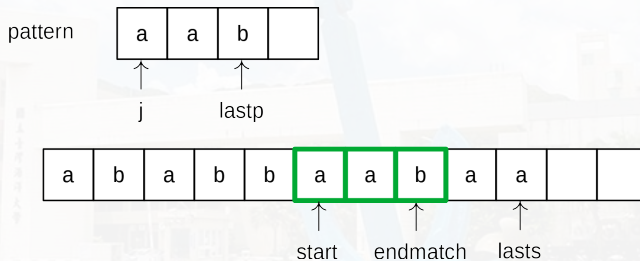












Time Complexity

- Worst case: $O(n \cdot m)$.

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- Actually, we can do much better for the pattern matching.



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Knuth, Morris, & Pratt (1977)

Home → SIAM Journal on Computing → Vol. 6, Iss. 2 (1977) → 10.1137/0206024

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Fast Pattern Matching in Strings

Authors: Donald E. Knuth, James H. Morris, Jr., and Vaughan R. Pratt | [AUTHORS INFO & AFFILIATIONS](#)
<https://doi.org/10.1137/0206024>

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Abstract

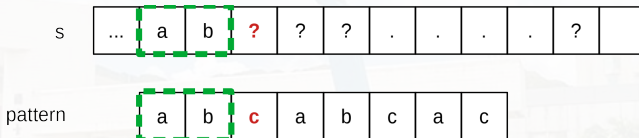
An algorithm is presented which finds all occurrences of one given string within another, in running time proportional to the sum of the lengths of the strings. The constant of proportionality is low enough to make this algorithm of practical use, and the procedure can also be extended to deal with some more general pattern-matching problems. A theoretical application of the algorithm shows that the set of concatenations of even palindromes, i.e., the language $\{\alpha\alpha^R\}^*$, can be recognized in linear time. Other algorithms which run even faster on the average are also considered.

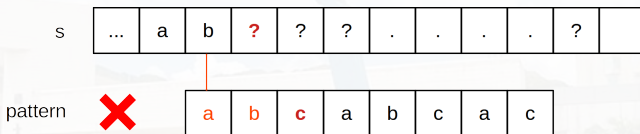
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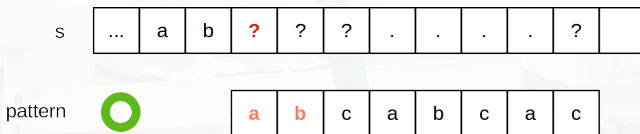
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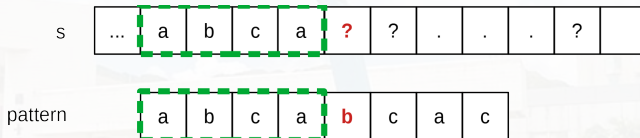
- *SIAM Journal on Computing*, Vol. 6(2) (1977), pp. 323–350.

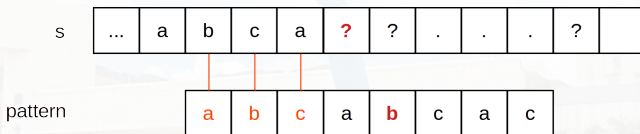


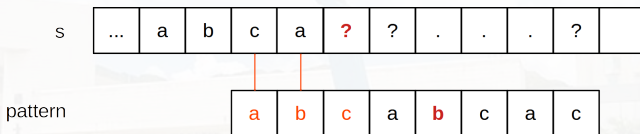


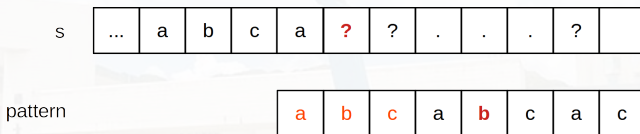


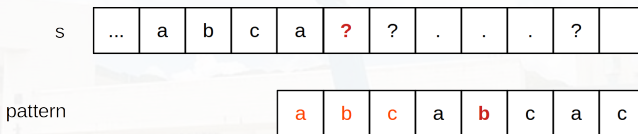












- We can “learn some information” from the pattern and the mismatches.

Key Observation

Observation

Suppose that

- $\text{Substr}(p, 0, k) = \text{Substr}(s, x, k)$ but
- $p[k] \neq s[x + k]$.

Then, for any $0 < r < k$, if $\text{Substr}(s, x + r, k - r)$ is **not** a *prefix* of p , we know p **cannot** occur as a substring of s at position $x + r$.



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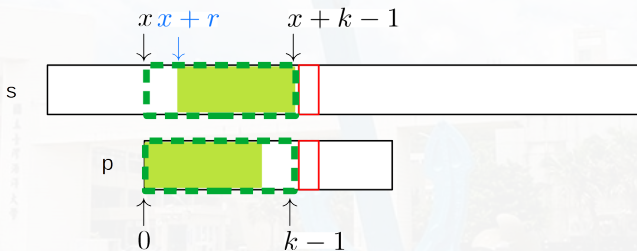
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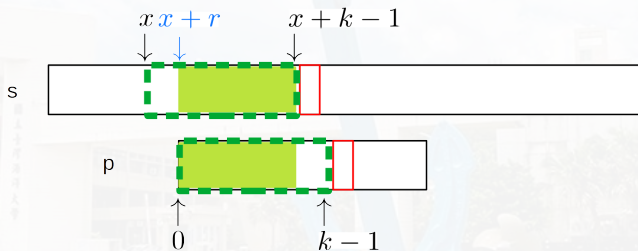
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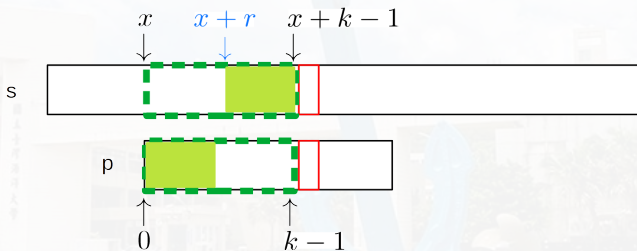
Then, for any $0 < r < k$, if $\text{Substr}(s, x + r, k - r)$ is **not** a *prefix* of p , we know p **cannot** occur as a substring of s at position $x + r$.

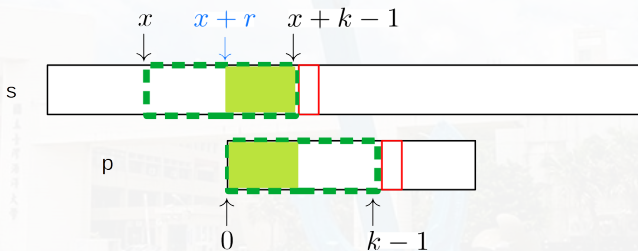
- We can bypass the impossibilities!











The failure function

If $p = p_0p_1 \cdots p_{m-1}$ is a pattern, then the failure function f is defined as $f(j) =$

- largest $k < j$ such that

$$p_0p_1 \cdots p_k = p_{j-k}p_{j-k+1} \cdots p_j$$

if such $k \geq 0$ exists

- -1 otherwise.



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if such $k \geq 0$ exists

- -1 otherwise.
- For example, suppose $p = \text{abcabcacab}$,

j	0	1	2	3	4	5	6	7	8	9
p	a	b	c	a	b	c	a	c	a	b
failure	-1	-1	-1	0	1	2	3	-1	0	1



The rule of using the failure function

- If a partial match is found and $j \neq 0$ such that

$$s_{i-j} \cdots s_{i-1} = p_0 p_1 \cdots p_{j-1}$$

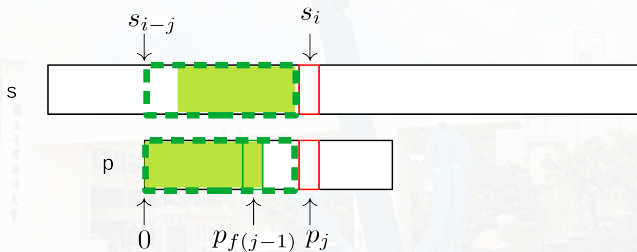
and

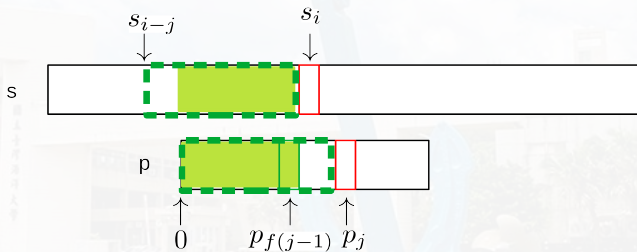
$$s_i \neq p_j,$$

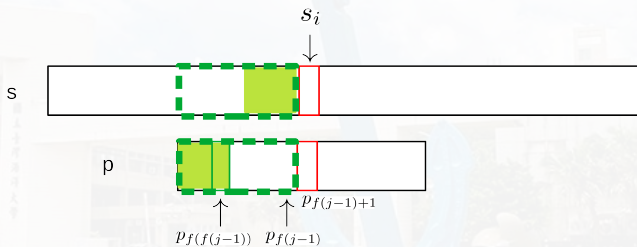
then matching may be resumed by comparing s_i and $p_{f(j-1)+1}$.

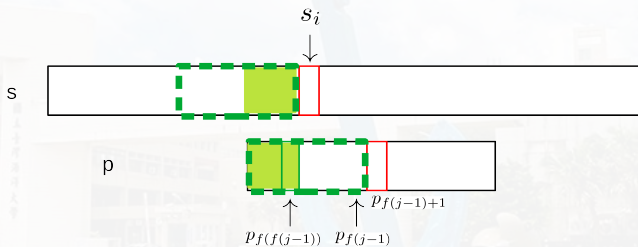
- If $j = 0$, then we may continue by comparing s_{i+1} and p_0 .

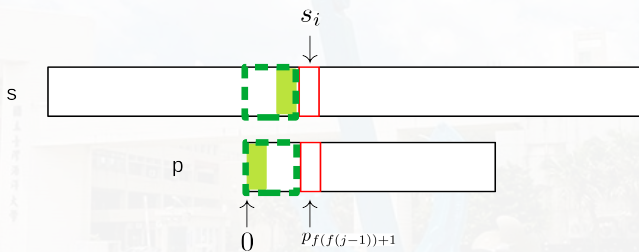












A fast way to compute the failure function

$$f(j) = \begin{cases} -1, & j = 0, \\ f^m(j-1) + 1, & \text{if } m \geq 0 \text{ is the least integer such that } p_{f^m(j-1)+1} = p_j, \\ -1, & \text{otherwise.} \end{cases}$$

- **Note:** $f^1(j) = f(j)$ and $f^m(j) = f(f^{m-1}(j))$.



Failure Function Code ($O(m)$)

```
void fail(const char* p, int* f) { // p: pattern; f: failure
    /* compute the pattern's failure function */
    int m = strlen(p);
    f[0] = -1;
    for (int j = 1; j < m; j++) {
        int i = f[j-1]; // i: prefix length
        while ((i >= 0) && (p[j] != p[i+1])) i = f[i];
        if (p[j] == p[i+1])
            f[j] = i + 1;
        else
            f[j] = -1;
    }
}
```

<i>j</i>	0	1	2	3	4	5	6	7	8	9
<i>p</i>	a	b	c	a	b	c	a	c	a	b
<i>f</i>	-1	-1	-1	0	1	2	3	-1	0	1



The time complexity of fail

- In each iteration of the **while** loop, the value of i decreases ($\because f$).
- i is reset at the beginning of each iteration of the for loop.
 - It is either reset to -1 or to a value 1 greater than its terminal value on the previous iteration (i.e., $f[j] = i + 1$).
 - Since the for loop is iterated only $m - 1$ times, the value of i is incremented for at most $m - 1$ times.
 - Hence, it cannot be decremented more than $m - 1$ times.
- The while loop is iterated $\leq m - 1$ times overall.



Declarations

```
#include <stdio.h>
#include <string.h>

#define MAX_STRING_SIZE 100
#define MAX_PATTERN_SIZE 100

int KMPmatch(void);
void fail(void);

int f[MAX_PATTERN_SIZE]; // computed by the failure function
char s[MAX_STRING_SIZE];
char p[MAX_PATTERN_SIZE];
```



The KMP Algorithm

```
int KMPmatch(char *s, char *p) {  
    int i = 0, j = 0; // i: for string s, and j: for pattern p  
    int n = strlen(s); // n: length of string s  
    int m = strlen(p); // m: length of pattern p  
  
    while (i < n && j < m) {  
        if (s[i] == p[j]) {  
            i++; j++;  
        } else if (j == 0) { // s[i] doesn't match p[0]  
            i++;  
        } else { // already having a matching substring  
            j = f[j-1] + 1; // the failure function acts here  
        }  
    }  
    return (j == m) ? (i - m) : -1;  
}
```

- the pattern is not found $\Rightarrow$ the pattern index $j \neq m \Rightarrow$ return -1 .
- the pattern is found $\Rightarrow$ the starting position is $i - m$.



Analysis of KMPmatch

- The while loop is iterated until the end of either the string or the pattern is reached.
- i is never decreased, so the lines that increase i cannot be executed $> n$ times.
- The resetting of j to $f[j - 1] + 1$ decreases the value of j .
 - This cannot be done more times than j is incremented by the statement $j++$ (otherwise j falls off the pattern).
- Each time $j++$ is executed, i is also incremented.
 - So, j cannot be incremented $> n$ times.
- No statement in the program is executed $> n$ times, therefore the time complexity of KMTmatch is $O(n + m)$ ($O(m)$ for `fail()`).



Discussions

